PRODUCT CASE STUDY · RAPIDO BIKE TAXI

Rapido Shield

Closing the trust gap for women at the three moments a bike-taxi ride can go wrong: pickup, the ride, and after the drop.

Anurag Tripathi

01 · CONTEXT

Why Rapido, and why now

6.5M

daily rides in March 2026, across bike, auto and cab

~35%

of rides in big cities are booked by women (co-founder, 2025)

$3B

valuation after a $240M round in May 2026

15.6

New MoRTH clause: women can choose a woman driver, where available

Rapido is winning on **price and speed**. The next constraint on growth is not demand. It is whether women, and regulators, **trust a stranger on a two-wheeler**.

Sources: Mint via Hyring (rides, Mar 2026) · Economic Times (funding) · YourStory (women share) · MoRTH amendment to Aggregator Guidelines 2025

02 · THE PROBLEM

Trust breaks at three moments, and today the rider carries the burden at each one

1 · Pickup

A captain arrives on a different bike from the one in the app. Rapido's advice: decline the ride. The rider has to spot the mismatch herself.

Bengaluru report

2 · The ride

Harassment and rash driving mid-ride. An SOS button exists, but pressing it while seated behind the person is the hardest moment to act.

Bengaluru; Assam, Feb 2026

3 · After drop

The captain now knows where she lives. Women have reported unwanted WhatsApp messages after the trip ended.

Ghaziabad, UP, Apr 2026

Problem statement: How might Rapido make every bike ride verifiably safe for women riders, without adding friction that costs captains rides or riders time?

Sources: Deccan Herald, Organiser, The Logical Indian, Asianet Newsable (reported incidents) · 03

03 · WHY IT MATTERS TO THE BUSINESS

For bike taxis, safety is not a feature. It is the licence to operate.

Regulatory risk	Karnataka halted bike taxis for months in 2025. The 2025 aggregator guidelines list failure to ensure passenger safety as grounds for suspension.
Demand ceiling	Every viral incident pushes women back to autos and cabs, which cost more. Night and first-ride usage is where fear shows up first.
Competitive moat	Uber already runs women-only Moto. With zero-commission models spreading, price stops differentiating. Trust can.

Sources: Outlook Business, Business Today (Karnataka); MediaNama (Guidelines 2025); YourStory (Uber Moto Women) · 04

04 · USERS

Two sides of the same ride

Hypotheses · validating with interviews

PRIMARY · RIDER

The late-shift commuter

Mid-20s, works till 9 PM in a metro. Picks a bike because an auto costs roughly double.

Job: get home fast and cheap without feeling exposed.

Fear: is this even the right person?

SECONDARY · RIDER

The tier-2 student

19, rides to coaching in a city like Kanpur or Lucknow. Parents decide whether she may use bike taxis.

Job: prove to family that the ride is safe.

Fear: her number and address are now known.

SUPPLY SIDE · CAPTAIN

The full-time captain

Early 30s, rides 10+ hours a day. Every minute lost at pickup is income lost.

Job: complete more rides per hour.

Fear: a false complaint or a bad model blocks his ID.

A safety product that ignores the captain fails twice: captains work around it, and supply drops at night when riders need it most.

05 · RIDER JOURNEY

Where today's safety features stop working

Stage	What exists today	The gap
Book	Captain photo, bike number	Details shown, but never checked against reality
Pickup	Ride OTP shared by rider	OTP proves the phone, not the person or the bike
Ride	SOS button, live location share	Needs the rider to act, at the moment she least can
Drop	Night call from safety team	Comes after the risk window, not during it
After	Number masking in the app	Captain has already seen her home address

Insight: every current feature is **reactive and rider-triggered**. The opportunity is to make the platform verify and watch, so she does not have to.

Existing features: YourStory, Deccan Herald, The CSR Journal · 06

06 · SOLUTION

Rapido Shield: one layer per moment

PICKUP · AI-ASSISTED

Verified Pickup

The ride cannot start until the captain's face and number plate match the app.

RIDE · AI FLAGS, HUMANS ACT

Ride Guardian

Spots route deviations and odd stops, then sends a silent check-in before a human steps in.

AFTER · NO AI NEEDED

Closed Loop

Contact channels shut at drop. Exact address hidden from the captain's trip history.

Baseline, not a bet: women-captain matching under MoRTH clause 15.6 ships for compliance, with honest "not available nearby" messaging.

07 · LAYER 1

Verified Pickup

The OTP proves the phone. This proves the person and the bike, before the ride can start.

Mismatch? The ride never starts. The rider gets a free priority rebook, and the captain gets a human review, not an automatic ban.

Part	Spec
Input	Captain selfie + photo of the number plate, taken in the captain app at pickup
AI does	Face match against onboarding KYC photo; plate OCR against the registered vehicle
Output	Match unlocks the OTP. Rider sees a "Verified" badge with a timestamp
Low confidence	Rider is shown the live photo and confirms herself. Never a silent pass
Friction budget	Under 15 seconds added per pickup, or it does not ship

08 · LAYER 2

Ride Guardian

Today she must press SOS. Guardian notices first, and asks quietly.

The check-in looks like a normal "Rate your route" prompt. One tap means fine; any other tap, or silence, alerts a human agent without tipping off the captain.

Part	Spec
Input	Live GPS, planned route, speed, stop duration, time of day
AI does	Anomaly score: off-route distance, unexplained stops, a destination change
Output	Silent check-in on the rider's phone. No answer in 45s escalates to the safety desk
Human step	Agent sees the live trip and calls the rider. Only a human calls police
Scope v1	Women riders, bike rides, 8 PM to 6 AM, opt-out available

09 · LAYER 3

Closed Loop

The cheapest fix in the deck, and it uses no AI at all.

Why no AI? The rule is simple and must never fail. A deterministic rule is cheaper, auditable and easy to explain to a regulator.

Rule	What changes
Expiring line	Masked calling and chat close 10 minutes after drop
Fuzzy drop	Captain's trip history shows the locality, not the exact pin
Landmark drop	Rider can set a drop point near, not at, her home
1-tap report	"Captain contacted me after the ride" goes straight to a priority queue
Consequence	Verified off-platform contact means suspension after human review

10 · HUMAN-IN-THE-LOOP DESIGN

AI makes the safety team faster and sharper

Moment	What AI does	Where a person steps in
SOS pressed	Hands the agent the live trip, route history and nearest help in one view	Every SOS reaches an agent instantly
Captain review	Spots patterns across complaints and trips that no one would see by hand	A human decides, with a 48-hour appeal
Ride audio	Summarises opt-in recordings so a review takes minutes, not hours	Recorded only with consent (DPDP Act 2023)
Captain coaching	Detects speeding and harsh braking, then nudges targeted training	Used to coach, never shown as a rider-facing score
Police escalation	Drafts the incident report with timeline and GPS trail	Agent confirms with the rider, then escalates

Principle: **AI widens what we notice and speeds up every response. People own the final call on a person.**

How it fails, and how we will know

Known failure modes

Night, helmets, cheap cameras lower face-match accuracy. Low confidence falls back to rider confirmation.

GPS drift in old-city lanes looks like a detour. Thresholds are tuned per city.

Legitimate detours such as closures or a rider's request. She can mark "I asked for this".

Dirty or non-standard plates defeat OCR. The fallback is a photo for the rider.

Evaluation plan

Offline: labelled pickup photos. Target false rejects under 2%, so honest captains are not blocked.

Shadow mode, 4 weeks: Guardian scores rides but shows nothing. Compare its flags with real complaints.

Online: share of check-ins that were real issues, and alert fatigue capped at 1 per rider per week.

RICE says: ship the boring fix first

Layer	Reach	Impact	Confid.	Effort	Score
1. Closed Loop	100%	2	80%	1.5	107
2. Verified Pickup	100%	3	70%	4	53
3.Women-captain match	10%	3	80%	2	12
4. Ride Guardian	30%	3	50%	6	8

Assumptions: Reach = share of women's bike rides touched. Impact on a 1–3 scale. Effort in person-months. Guardian covers night rides only, roughly 30% (assumed).

Override: women-captain matching scores low but ships anyway, since clause 15.6 makes it mandatory. Guardian is last because it needs shadow-mode data first.

13 · SUCCESS METRICS

Trust shows up as women coming back

NORTH STAR

7-day repeat rate of women riders on bike taxis, split by day and night

Surveys measure what people say. Coming back after dark measures what they believe.

Input metrics

- Mismatched pickups blocked per 10k rides
- Post-ride contact reports per 100k rides
- Median time from flag to human call
- Share of check-ins that were real issues

Guardrails (must not get worse)

- Pickup time added by verification
- Captain rides per hour and weekly churn
- Share of penalties overturned on appeal
- Ride cancellations after captain arrival

14 · LAUNCH PLAN

Earn the right to scale, one gate at a time

WEEKS 0–2

Discover

5 rider and 5 captain interviews. Review safety tickets by stage of ride.

Gate: the 3 moments hold up in the data

WEEKS 3–6

Closed Loop

Ship nationwide behind a flag. Low risk, no model.

Gate: contact reports fall; no rise in failed drops

WEEKS 6–14

Verified Pickup

One city, night rides, A/B test by zone. Captains briefed first.

Gate: under 15s added; captain churn flat

WEEKS 10–20

Ride Guardian

4 weeks in shadow mode, then live for opted-in riders.

Gate: alert precision meets target

Kill criteria are written before launch: if a gate fails twice, the layer goes back to discovery instead of being pushed through.

AI watches every ride. People make the calls that matter.

Great AI product work means putting the model where it adds the most value, verifying at pickup and detecting risk mid-ride, then launching it safely with shadow mode, clear metrics and a human in the loop.

Anurag Tripathi

What would you test first? I'd love to hear from people who build mobility or trust and safety products.